Mary Ogunmola

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EDUCATION

Ph.D. **University of Utah**, Kahlert School of Computing
Computer Science (expected 2030)
Advisor: Professor Robert Ricci
CGPA: 4.00/4.00
Coursework: Software and Systems Security; Secure Computing; Operating Systems; Computer Architecture; Parallel Computing and HPC.

B.Sc. **Joseph Ayo Babalola University**
Computer Science (2023)
CGPA: 4.89/5.00 (Valedictorian)

RESEARCH INTERESTS

Verifiable Trust and Security, Confidential Computing, Systems Performance, and Reproducibility.

RESEARCH EXPERIENCE

2025 – Present **Graduate Research Assistant**, Flux Research Group, University of Utah
Advisor: Professor Robert Ricci

Secure deployment for cloud-hosted 5G core networks
- Designed a framework for the secure execution of 5G core network functions on untrusted cloud infrastructure by running Open5GS network functions inside Trusted Execution ENnironments.
- Extended the framework with remote attestation to detect runtime integrity changes in deployed network functions.

Performance reproducibility and hardware sensitivity
- Evaluating systems artifacts on original-equivalent and different hardware environments to assess how well results reproduce.
- Developing an experimentation tool that deploys systems experiments across heterogeneous nodes, helping researchers identify sensitivity before drawing general conclusions.

2023 – 2024 **Undergraduate Research Assistant**, Joseph Ayo Babalola University
Advisor: Dr. V. A. Olutayo
Analyzed cloud-centric IoT transformations through an enterprise case study, characterizing cost, performance, and security trade-offs to inform migration strategies. Designed and evaluated a unified academic resource system, demonstrating over 120% improvement in accessibility.

PROJECTS

Container Isolation Security Evaluation *(Docker, Kata Containers, Linux)*
Reproduced published container-escape CVEs in controlled environments to compare the containment provided by Docker's shared-kernel model and Kata Containers' VM-based isolation, documenting whether each attack crossed container, kernel, or VM boundaries.

Insider-Threat Anomaly Detection *(CERT Insider Threat Dataset, Python, Isolation Forest, XGBoost)*
Built an ensemble detection pipeline combining rule-based, unsupervised, and supervised signals into behavioral risk scores over time windows.

ACADEMIC SERVICE

2026 Artifact Evaluation Committee: ACM CCS 2026; PoPETs 2026; OSDI 2026.

2026 Shadow Program Committee: ACM SIGMETRICS 2026.

2026 Judge, University of Utah Senior Capstone Project Showcase.

2025 Poster Evaluator, University of Utah Undergraduate Research Symposium.

HONORS, AWARDS & FELLOWSHIPS

2025 University of Utah Graduate Fellowship.

2024 EducationUSA Opportunity Funds Award.

2023 Overall Best Graduating Student (Valedictorian), Joseph Ayo Babalola University; Dean's Prize; University Scholar; Pro Chancellor's Prize; Parent Forum's Prize; Most Outstanding Student (Nigerian Association of Computing Students).

PROFESSIONAL EXPERIENCE

2024 – 2025 **Software Engineer**, i-Vantage Technology, Lagos, Nigeria
Maintained AWS-hosted production services and integrated vulnerability scanning into deployment workflows; developed a CRM platform serving mortgage and real-estate organizations.

2024 **Cybersecurity Analyst**, Hacktify.

2022 **Software Developer Intern**, JIT Solutions (Remote)
Developed reusable React and React Native components and delivered technical training to entry-level developers.

TEACHING EXPERIENCE

2021 – 2023 **Tutor**, Nigerian Association of Computing Students, Joseph Ayo Babalola University *(CS 314, CS 220)*.

TALKS & PRESENTATIONS

Oct 2026 Panelist, Women in Computer Science, Brigham Young University: "CS Graduate Student Panel"

Jun 2023 Presenter, Joseph Ayo Babalola University Seminar: "Improving the Accessibility of Academic Resources in Educational Institutions."

TECHNICAL SKILLS

Trusted Computing: Intel SGX; Intel TDX; Gramine; Keylime; Linux IMA; Kata Containers; confidential containers.

Systems: Linux; QEMU/KVM; Docker; Kubernetes; xv6.

Measurement: perf; eBPF/bpftrace; ftrace; GDB; tcpdump; experimental design; statistical analysis.

Experimentation: CloudLab; Nix; reproducible experiment design; artifact packaging and evaluation.

Networking: 5G core networks; Open5GS; UERANSIM

Cloud: AWS; Azure; GCP; Terraform; CI/CD.

Programming: C; C++; Python; Bash; JavaScript; OpenMP.

LEADERSHIP & MENTORSHIP

2025 – Present **Co-Chair**, Grad-CS Women, Kahlert School of Computing, University of Utah.

2025 – Present **Secretary**, Graduate Student Advisory Council (GradSAC), Kahlert School of Computing, University of Utah.

2025 **Mentor**, EducationUSA Opportunity Funds Program Scholars, Cohort 2025. Advised international applicants through graduate school applications.

2024 **Community Organizer**, AWS Nigeria User Group.

2022 – 2023 **Team Leader**, Academic Committee, Nigerian Association of Computing Students (NACOS), JABU Chapter.

PROFESSIONAL AFFILIATIONS

Association for Computing Machinery (ACM); IEEE Computer Society; National Society of Black Engineers; Society of Women Engineers.